

MATHEMATICAL BACKGROUND

A.1 Complexity Analysis and $O()$ Notation

Benchmarking

Analysis of
algorithms

Computer scientists are often faced with the task of comparing algorithms to see how fast they run or how much memory they require. There are two approaches to this task. The first is **benchmarking**—running the algorithms on a computer and measuring speed in seconds and memory consumption in bytes. Ultimately, this is what really matters, but a benchmark can be unsatisfactory because it is so specific: it measures the performance of a particular program written in a particular language, running on a particular computer, with a particular compiler and particular input data. From the single result that the benchmark provides, it can be difficult to predict how well the algorithm would do on a different compiler, computer, or data set. The second approach relies on a mathematical **analysis of algorithms**, independent of the particular implementation and input, as discussed below.

A.1.1 Asymptotic analysis

We will consider algorithm analysis through the following example, a program to compute the sum of a sequence of numbers:

```
function SUMMATION(sequence) returns a number
    sum ← 0
    for  $i = 1$  to LENGTH(sequence) do
        sum ← sum + sequence[i]
    return sum
```

The first step in the analysis is to abstract over the input, in order to find some parameter or parameters that characterize the size of the input. In this example, the input can be characterized by the length of the sequence, which we will call n . The second step is to abstract over the implementation, to find some measure that reflects the running time of the algorithm but is not tied to a particular compiler or computer. For the SUMMATION program, this could be just the number of lines of code executed, or it could be more detailed, measuring the number of additions, assignments, array references, and branches executed by the algorithm. Either way gives us a characterization of the total number of steps taken by the algorithm as a function of the size of the input. We will call this characterization $T(n)$. If we count lines of code, we have $T(n) = 2n + 2$ for our example.

If all programs were as simple as SUMMATION, the analysis of algorithms would be a trivial field. But two problems make it more complicated. First, it is rare to find a parameter like n that completely characterizes the number of steps taken by an algorithm. Instead, the best we can usually do is compute the worst case $T_{\text{worst}}(n)$ or the average case $T_{\text{avg}}(n)$. Computing an average means that the analyst must assume some distribution of inputs.

The second problem is that algorithms tend to resist exact analysis. In that case, it is necessary to fall back on an approximation. We say that the SUMMATION algorithm is $O(n)$, meaning that its measure is at most a constant times n , with the possible exception of a few small values of n . More formally,

$T(n)$ is $O(f(n))$ if $T(n) \leq kf(n)$ for some k , for all $n > n_0$.

The $O()$ notation gives us what is called an **asymptotic analysis**. We can say without question that, as n asymptotically approaches infinity, an $O(n)$ algorithm is better than an $O(n^2)$ algorithm. A single benchmark figure could not substantiate such a claim.

Asymptotic analysis

The $O()$ notation abstracts over constant factors, which makes it easier to use, but less precise, than the $T()$ notation. For example, an $O(n^2)$ algorithm will always be worse than an $O(n)$ in the long run, but if the two algorithms are $T(n^2 + 1)$ and $T(100n + 1000)$, then the $O(n^2)$ algorithm is actually better for $n < 110$.

Despite this drawback, asymptotic analysis is the most widely used tool for analyzing algorithms. It is precisely because the analysis abstracts over both the exact number of operations (by ignoring the constant factor k) and the exact content of the input (by considering only its size n) that the analysis becomes mathematically feasible. The $O()$ notation is a good compromise between precision and ease of analysis.

A.1.2 NP and inherently hard problems

The analysis of algorithms and the $O()$ notation allow us to talk about the efficiency of a particular algorithm. However, they have nothing to say about whether there could be a better algorithm for the problem at hand. The field of **complexity analysis** analyzes problems rather than algorithms. The first gross division is between problems that can be solved in polynomial time and problems that cannot be solved in polynomial time, no matter what algorithm is used. The class of polynomial problems—those which can be solved in time $O(n^k)$ for some k —is called **P**. These are sometimes called “easy” problems, because the class contains those problems with running times like $O(\log n)$ and $O(n)$. But it also contains those with time $O(n^{1000})$, so the name “easy” should not be taken too literally.

Complexity analysis

P

Another important class of problems is **NP**, the class of nondeterministic polynomial problems. A problem is in this class if there is some algorithm that can guess a solution and then verify whether a guess is correct in polynomial time. The idea is that if you have an arbitrarily large number of processors so that you can try all the guesses at once, or if you are very lucky and always guess right the first time, then the NP problems become P problems. One of the biggest open questions in computer science is whether the class NP is equivalent to the class P when one does not have the luxury of an infinite number of processors or omniscient guessing. Most computer scientists are convinced that $P \neq NP$; that NP problems are inherently hard and have no polynomial-time algorithms. But this has never been proven.

NP

Those who are interested in deciding whether $P = NP$ look at a subclass of NP called the **NP-complete** problems. The word “complete” is used here in the sense of “most extreme” and thus refers to the hardest problems in the class NP. It has been proven that either all the NP-complete problems are in P or none of them is. This makes the class theoretically interesting, but the class is also of practical interest because many important problems are known to be NP-complete. An example is the satisfiability problem: given a sentence of propositional logic, is there an assignment of truth values to the proposition symbols of the sentence that makes it true? Unless a miracle occurs and $P = NP$, there can be no algorithm

NP-complete

that solves *all* satisfiability problems in polynomial time. However, AI is more interested in whether there are algorithms that perform efficiently on *typical* problems drawn from a predetermined distribution; as we saw in Chapter 7, there are algorithms such as WALKSAT that do quite well on many problems.

NP-hard

The class of **NP-hard** problems consists of those problems that are reducible (in polynomial time) to all the problems in NP, so if you solved any NP-hard problem, you could solve all the problems in NP. The NP-complete problems are all NP-hard, but there are some NP-hard problems that are even harder than NP-complete.

Co-NP

The class **co-NP** is the complement of NP, in the sense that, for every decision problem in NP, there is a corresponding problem in co-NP with the “yes” and “no” answers reversed. We know that P is a subset of both NP and co-NP, and it is believed that there are problems in co-NP that are not in P. The **co-NP-complete** problems are the hardest problems in co-NP.

Co-NP-complete

The class **#P** (pronounced “number P” according to Garey and Johnson (1979), but often pronounced “sharp P”) is the set of counting problems corresponding to the decision problems in NP. Decision problems have a yes-or-no answer: is there a solution to this 3-SAT formula? Counting problems have an integer answer: how many solutions are there to this 3-SAT formula? In some cases, the counting problem is much harder than the decision problem. For example, deciding whether a bipartite graph has a perfect matching can be done in time $O(VE)$ (where the graph has V vertices and E edges), but the counting problem “how many perfect matches does this bipartite graph have” is #P-complete, meaning that it is hard as any problem in #P and thus at least as hard as any NP problem.

Another class is the class of PSPACE problems—those that require a polynomial amount of space, even on a nondeterministic machine. It is believed that PSPACE-hard problems are worse than NP-complete problems, although it could turn out that $NP = PSPACE$, just as it could turn out that $P = NP$.

A.2 Vectors, Matrices, and Linear Algebra

Vector

Mathematicians define a **vector** as a member of a vector space, but we will use a more concrete definition: a vector is an ordered sequence of values. For example, in two-dimensional space, we have vectors such as $\mathbf{x} = \langle 3, 4 \rangle$ and $\mathbf{y} = \langle 0, 2 \rangle$. We follow the convention of boldface characters for vector names, although some authors use arrows or bars over the names: $\vec{x}$ or $\bar{y}$. The elements of a vector can be accessed using subscripts: $\mathbf{z} = \langle z_1, z_2, \dots, z_n \rangle$. One confusing point: this book is synthesizing work from many subfields, which variously call their sequences vectors, lists, or tuples, and variously use the notations $\langle 1, 2 \rangle$, $[1, 2]$, or $(1, 2)$.

The two fundamental operations on vectors are vector addition and scalar multiplication. The vector addition $\mathbf{x} + \mathbf{y}$ is the elementwise sum: $\mathbf{x} + \mathbf{y} = \langle 3 + 0, 4 + 2 \rangle = \langle 3, 6 \rangle$. Scalar multiplication multiplies each element by a constant: $5\mathbf{x} = \langle 5 \times 3, 5 \times 4 \rangle = \langle 15, 20 \rangle$.

The length of a vector is denoted $|\mathbf{x}|$ and is computed by taking the square root of the sum of the squares of the elements: $|\mathbf{x}| = \sqrt{(3^2 + 4^2)} = 5$. The dot product $\mathbf{x} \cdot \mathbf{y}$ (also called scalar product) of two vectors is the sum of the products of corresponding elements, that is, $\mathbf{x} \cdot \mathbf{y} = \sum_i x_i y_i$, or in our particular case, $\mathbf{x} \cdot \mathbf{y} = 3 \times 0 + 4 \times 2 = 8$.

Vectors are often interpreted as directed line segments (arrows) in an n -dimensional Euclidean space. Vector addition is then equivalent to placing the tail of one vector at the head of the other, and the dot product $\mathbf{x} \cdot \mathbf{y}$ is $|\mathbf{x}| |\mathbf{y}| \cos \theta$, where θ is the angle between $\mathbf{x}$ and $\mathbf{y}$.

A **matrix** is a rectangular array of values arranged into rows and columns. Here is a [Matrix](#) $\mathbf{A}$ of size 3×4 :

$$\begin{pmatrix} \mathbf{A}_{1,1} & \mathbf{A}_{1,2} & \mathbf{A}_{1,3} & \mathbf{A}_{1,4} \\ \mathbf{A}_{2,1} & \mathbf{A}_{2,2} & \mathbf{A}_{2,3} & \mathbf{A}_{2,4} \\ \mathbf{A}_{3,1} & \mathbf{A}_{3,2} & \mathbf{A}_{3,3} & \mathbf{A}_{3,4} \end{pmatrix}$$

The first index of $\mathbf{A}_{i,j}$ specifies the row and the second the column. In programming languages, $\mathbf{A}_{i,j}$ is often written $\mathbf{A}[i, j]$ or $\mathbf{A}[i][j]$.

The sum of two matrices is defined by adding their corresponding elements; for example $(\mathbf{A} + \mathbf{B})_{i,j} = \mathbf{A}_{i,j} + \mathbf{B}_{i,j}$. (The sum is undefined if $\mathbf{A}$ and $\mathbf{B}$ have different sizes.) We can also define the multiplication of a matrix by a scalar: $(c\mathbf{A})_{i,j} = c\mathbf{A}_{i,j}$. Matrix multiplication (the product of two matrices) is more complicated. The product $\mathbf{AB}$ is defined only if $\mathbf{A}$ is of size $a \times b$ and $\mathbf{B}$ is of size $b \times c$ (i.e., the second matrix has the same number of rows as the first has columns); the result is a matrix of size $a \times c$. If the matrices are of appropriate size, then the result is

$$(\mathbf{AB})_{i,k} = \sum_j \mathbf{A}_{i,j} \mathbf{B}_{j,k}.$$

Matrix multiplication is not commutative, even for square matrices: $\mathbf{AB} \neq \mathbf{BA}$ in general. It is, however, associative: $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$. Note that the dot product can be expressed in terms of a transpose and a matrix multiplication: $\mathbf{x} \cdot \mathbf{y} = \mathbf{x}^\top \mathbf{y}$.

The **identity matrix** $\mathbf{I}$ has elements $\mathbf{I}_{i,j}$ equal to 1 when $i = j$ and equal to 0 otherwise. It has the property that $\mathbf{AI} = \mathbf{A}$ for all $\mathbf{A}$. The **transpose** of $\mathbf{A}$, written $\mathbf{A}^\top$ is formed by turning rows into columns and vice versa, or, more formally, by $\mathbf{A}^\top_{i,j} = \mathbf{A}_{j,i}$. The **inverse** of a square matrix $\mathbf{A}$ is another square matrix $\mathbf{A}^{-1}$ such that $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$. For a **singular** matrix, the inverse does not exist. For a nonsingular matrix, it can be computed in $O(n^3)$ time. [Identity matrix](#) [Transpose](#) [Inverse](#) [Singular](#)

Matrices are used to solve systems of linear equations in $O(n^3)$ time; the time is dominated by inverting a matrix of coefficients. Consider the following set of equations, for which we want a solution in x , y , and z :

$$\begin{aligned} +2x + y - z &= 8 \\ -3x - y + 2z &= -11 \\ -2x + y + 2z &= -3. \end{aligned}$$

We can represent this system as the matrix equation $\mathbf{Ax} = \mathbf{b}$, where

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & -1 \\ -3 & -1 & 2 \\ -2 & 1 & 2 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 8 \\ -11 \\ -3 \end{pmatrix}.$$

To solve $\mathbf{Ax} = \mathbf{b}$ we multiply both sides by $\mathbf{A}^{-1}$, yielding $\mathbf{A}^{-1}\mathbf{Ax} = \mathbf{A}^{-1}\mathbf{b}$, which simplifies to $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$. After inverting $\mathbf{A}$ and multiplying by $\mathbf{b}$, we get the answer

$$\mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}.$$

A few more miscellaneous points: we use $\log(x)$ for the natural logarithm, $\log_e(x)$. We use $\operatorname{argmax}_x f(x)$ for the value of x for which $f(x)$ is maximal.

A.3 Probability Distributions

A probability is a measure over a set of events that satisfies three axioms:

1. The measure of each event is between 0 and 1. We write this as $0 \leq P(X=x_i) \leq 1$, where X is a random variable representing an event and x_i are the possible values of X . In general, random variables are denoted by uppercase letters and their values by lowercase letters.
2. The measure of the whole set is 1; that is, $\sum_{i=1}^n P(X=x_i) = 1$.
3. The probability of a union of disjoint events is the sum of the probabilities of the individual events; that is, $P(X=x_1 \vee X=x_2) = P(X=x_1) + P(X=x_2)$, in the case where x_1 and x_2 are disjoint.

A **probabilistic model** consists of a sample space of mutually exclusive possible outcomes, together with a probability measure for each outcome. For example, in a model of the weather tomorrow, the outcomes might be *sun*, *cloud*, *rain*, and *snow*. A subset of these outcomes constitutes an event. For example, the event of precipitation is the subset consisting of $\{\text{rain}, \text{snow}\}$.

We use $\mathbf{P}(X)$ to denote the vector of values $\langle P(X=x_1), \dots, P(X=x_n) \rangle$. We also use $P(x_i)$ as an abbreviation for $P(X=x_i)$ and $\sum_x P(x)$ for $\sum_{i=1}^n P(X=x_i)$.

The conditional probability $P(B|A)$ is defined as $P(B \cap A)/P(A)$. A and B are conditionally independent if $P(B|A) = P(B)$ (or equivalently, $P(A|B) = P(A)$).

For continuous variables, there are an infinite number of values, and unless there are point spikes, the probability of any one exact value is 0. So it makes more sense to talk about the value being within a range. We do that with a **probability density function**, which has a slightly different meaning from the discrete probability function. Since $P(X=x)$ —the probability that X has the value x exactly—is zero, we instead measure how likely it is that X falls into an interval around x , compared to the width of the interval, and take the limit as the interval width goes to zero:

$$P(x) = \lim_{dx \rightarrow 0} P(x \leq X \leq x + dx) / dx.$$

The density function must be nonnegative for all x and must have

$$\int_{-\infty}^{\infty} P(x) dx = 1.$$

We can also define the **cumulative distribution** $F_X(x)$, which is the probability of a random variable being less than x :

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x P(u) du.$$

Note that the probability density function has units, whereas the discrete probability function is unitless. For example, if values of X are measured in seconds, then the density is measured in Hz (i.e., 1/sec). If values of $\mathbf{X}$ are points in three-dimensional space measured in meters, then density is measured in $1/m^3$.

One of the most important probability distributions is the **Gaussian distribution**, also known as the **normal distribution**. We use the notation $\mathcal{N}(x; \mu, \sigma^2)$ for the normal distribu-

Probability density
function

Cumulative
distribution

Gaussian distribution

tion that is a function of x with mean μ and standard deviation σ (and therefore variance σ^2). It is defined as

$$\mathcal{N}(x; \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)},$$

where x is a continuous variable ranging from $-\infty$ to $+\infty$. With mean $\mu=0$ and variance $\sigma^2=1$, we get the special case of the **standard normal distribution**. For a distribution over a vector $\mathbf{x}$ in n dimensions, there is the **multivariate Gaussian** distribution:

Standard normal
distribution
Multivariate
Gaussian

$$\mathcal{N}(\mathbf{x}; \mu, \Sigma) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} e^{-\frac{1}{2} (\mathbf{x}-\mu)^\top \Sigma^{-1} (\mathbf{x}-\mu)},$$

where μ is the mean vector and Σ is the **covariance matrix** (see below). The cumulative distribution for a univariate normal distribution is given by

$$F(x) = \int_{-\infty}^x \mathcal{N}(z; \mu, \sigma^2) dz = \frac{1}{2} \left(1 + \operatorname{erf}\left(\frac{x-\mu}{\sigma\sqrt{2}}\right) \right),$$

where $\operatorname{erf}(x)$ is the so-called **error function**, which has no closed-form representation.

The **central limit theorem** states that the distribution formed by sampling n independent random variables and taking their mean tends to a normal distribution as n tends to infinity. This holds for almost any collection of random variables, even if they are not strictly independent, unless the variance of any finite subset of variables dominates the others.

Central limit
theorem

The **expectation** of a random variable, $E(X)$, is the mean or average value, weighted by the probability of each value. For a discrete variable it is:

Expectation

$$E(X) = \sum_i x_i P(X=x_i).$$

For a continuous variable, replace the summation with an integral and use the probability density function, $P(x)$:

$$E(X) = \int_{-\infty}^{\infty} x P(x) dx.$$

For any function f , we also have

$$E(f(X)) = \int_{-\infty}^{\infty} f(x) P(x) dx.$$

Finally, when necessary, one may specify the distribution for the random variable as a subscript to the expectation operator:

$$E_{X \sim Q(x)}(g(X)) = \int_{-\infty}^{\infty} g(x) Q(x) dx.$$

Besides the expectation, other important statistical properties of a distribution include the **variance**, which is the expected value of the square of the difference from the mean, μ , of the distribution:

Variance

$$\operatorname{Var}(X) = E((X - \mu)^2)$$

and the **standard deviation**, which is the square root of the variance.

Standard deviation

The **root mean square (RMS)** of a set of values (often samples of a random variable) is the square root of the mean of the squares of the values,

$$RMS(x_1, \dots, x_n) = \sqrt{\frac{x_1^2 + \dots + x_n^2}{n}}.$$

Covariance

The **covariance** of two random variables is the expectation of the product of their differences from their means:

$$\text{cov}(X, Y) = E((X - \mu_X)(Y - \mu_Y)).$$

Covariance matrix

The **covariance matrix**, often denoted Σ , is a matrix of covariances between elements of a vector of random variables. Given $\mathbf{X} = \langle X_1, \dots, X_n \rangle^\top$, the entries of the covariance matrix are as follows:

$$\Sigma_{i,j} = \text{cov}(X_i, X_j) = E((X_i - \mu_i)(X_j - \mu_j)).$$

Sampling

We say we **sample** from a probability distribution, when we pick a value at random. We don't know what any one pick will bring, but in the limit a large collection of samples will approach the same probability density function as the distribution it is sampled from. The **uniform distribution** is one where every element is equally (uniformly) probable. So when we say we “sample uniformly (at random) from the integers 0 to 99” it means that we are equally likely to pick any integer in that range.

Uniform distribution

Bibliographical and Historical Notes

The $O()$ notation so widely used in computer science today was first introduced in the context of number theory by the mathematician P. G. H. Bachmann (1894). The concept of NP-completeness was invented by Cook (1971), and the modern method for establishing a reduction from one problem to another is due to Karp (1972). Cook and Karp have both won the Turing award for their work.

Textbooks on the analysis and design of algorithms include Sedgewick and Wayne (2011) and Cormen, Leiserson, Rivest and Stein (2009). These books place an emphasis on designing and analyzing algorithms to solve tractable problems. For the theory of NP-completeness and other forms of intractability, see Garey and Johnson (1979) or Papadimitriou (1994). Good texts on probability include Chung (1979), Ross (2015), and Bertsekas and Tsitsiklis (2008).

NOTES ON LANGUAGES AND ALGORITHMS

B.1 Defining Languages with Backus–Naur Form (BNF)

In this book we define several languages, including the languages of propositional logic (235), first-order logic (276), and a subset of English (page 893). A formal language is defined as a set of strings where each string is a sequence of symbols. The languages we are interested in consist of an infinite set of strings, so we need a concise way to characterize the set. We do that with a **grammar**. The particular type of grammar we use is called a **context-free grammar**, because each expression has the same form in any context. We write our grammars in a formalism called **Backus–Naur form (BNF)**. There are four components to a BNF grammar:

Context-free
grammar
Backus–Naur form
(BNF)

- A set of **terminal symbols**. These are the symbols or words that make up the strings of the language. They could be letters (**A, B, C, ...**) or words (**a, aardvark, abacus, ...**), or whatever symbols are appropriate for the domain. Terminal symbol
- A set of **nonterminal symbols** that categorize subphrases of the language. For example, the nonterminal symbol *NounPhrase* in English denotes an infinite set of strings including “you” and “the big slobbery dog.” Nonterminal symbol
- A **start symbol**, which is the nonterminal symbol that denotes the complete set of strings of the language. In English, this is *Sentence*; for arithmetic, it might be *Expr*, and for programming languages it is *Program*. Start symbol
- A set of **rewrite rules**, of the form $LHS \rightarrow RHS$, where *LHS* is a nonterminal symbol and *RHS* is a sequence of zero or more symbols. These can be either terminal or nonterminal symbols, or the symbol ϵ , which is used to denote the empty string. Rewrite rules

A rewrite rule of the form

$Sentence \rightarrow NounPhrase VerbPhrase$

means that whenever we have two strings categorized as a *NounPhrase* and a *VerbPhrase*, we can append them together and categorize the result as a *Sentence*. As an abbreviation, the two rules ($S \rightarrow A$) and ($S \rightarrow B$) can be written ($S \rightarrow A \mid B$). To illustrate these concepts, here is a BNF grammar for simple arithmetic expressions:

$Expr \rightarrow Expr \text{ Operator } Expr \mid (Expr) \mid Number$

$Number \rightarrow Digit \mid Number Digit$

$Digit \rightarrow 0 \mid 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9$

$Operator \rightarrow + \mid - \mid \div \mid \times$

We cover languages and grammars in more detail in Chapter 24. Be aware that other books use slightly different notations for BNF; for example, you might see $\langle Digit \rangle$ instead of *Digit* for a nonterminal, ‘word’ instead of **word** for a terminal, or $: =$ instead of $\rightarrow$ in a rule.

B.2 Describing Algorithms with Pseudocode

Pseudocode

The algorithms in this book are described in **pseudocode**. Most of the pseudocode should be familiar to programmers who use languages like Java, C++, or especially Python. In some places we use mathematical formulas or ordinary English to describe parts that would otherwise be more cumbersome. A few idiosyncrasies should be noted:

- **Persistent variables:** We use the keyword **persistent** to say that a variable is given an initial value the first time a function is called and retains that value (or the value given to it by a subsequent assignment statement) on all subsequent calls to the function. Thus, persistent variables are like global variables in that they outlive a single call to their function; but they are accessible only within the function. The agent programs in the book use persistent variables for *memory*. Programs with persistent variables can be implemented as *objects* in object-oriented languages such as C++, Java, Python, and Smalltalk. In functional languages, they can be implemented by *functional closures* over an environment containing the required variables.
- **Functions as values:** Functions have capitalized names, and variables have lowercase italic names. So most of the time, a function call looks like $FN(x)$. However, we allow the value of a variable to be a function; for example, if the value of the variable f is the square root function, then $f(9)$ returns 3.
- **Indentation is significant:** Indentation is used to mark the scope of a loop or conditional, as in the languages Python and CoffeeScript, and unlike Java, C++, and Go (which use braces) or Lua and Ruby (which use **end**).
- **Destructuring assignment:** The notation “ $x, y \leftarrow pair$ ” means that the right-hand side must evaluate to a two-element collection, and the first element is assigned to x and the second to y . The same idea is used in “**for** x, y **in** *pairs* **do**” and can be used to swap two variables: “ $x, y \leftarrow y, x$ ”
- **Default values for parameters:** The notation “**function** $F(x, y=0)$ **returns** a number” means that y is an optional argument with default value 0; that is, the calls $F(3, 0)$ and $F(3)$ are equivalent.
- **yield:** a function that contains the keyword **yield** is a **generator** that generates a sequence of values, one each time the **yield** expression is encountered. After yielding, the function continues execution with the next statement. The languages Python, Ruby, C#, and Javascript (ECMAScript) have this same feature.
- **Loops:** There are four kinds of loops:
 - “**for** x **in** c **do**” executes the loop with the variable x bound to successive elements of the collection c .
 - “**for** $i = 1$ **to** n **do**” executes the loop with i bound to successive integers from 1 to n inclusive.
 - “**while** *condition* **do**” means the condition is evaluated before each iteration of the loop, and the loop exits if the condition is false.

Generator

- “**repeat ... until** *condition*” means that the loop is executed unconditionally the first time, then the condition is evaluated, and the loop exits if the condition is true; otherwise the loop keeps executing (and testing at the end).
- **Lists:** $[x, y, z]$ denotes a list of three elements. The “+” operator concatenates lists: $[1, 2] + [3, 4] = [1, 2, 3, 4]$. A list can be used as a stack: POP removes and returns the last element of a list, TOP returns the last element.
- **Sets:** $\{x, y, z\}$ denotes a set of three elements. $\{x : p(x)\}$ denotes the set of all elements x for which $p(x)$ is true.
- **Arrays start at 1:** the first index of an array is 1 as in usual mathematical notation (and in R and Julia), not 0 (as in Python and Java and C).

B.3 Online Supplemental Material

The book has a Web site with supplemental material, instructions for sending suggestions, and opportunities for joining discussion lists:

- aima.cs.berkeley.edu

The algorithms in the book, and multiple additional programming exercises, have been implemented in Python and Java (and some in other languages) at the online code repository, accessible from the Web site and currently hosted at:

- github.com/aimacode

Bibliography

The following abbreviations are used for frequently cited conferences and journals:

AAAI	Proceedings of the AAAI Conference on Artificial Intelligence
AAMAS	Proceedings of the International Conference on Autonomous Agents and Multi-agent Systems
ACL	Proceedings of the Annual Meeting of the Association for Computational Linguistics
AIJ	Artificial Intelligence (Journal)
AIMag	AI Magazine
AIPS	Proceedings of the International Conference on AI Planning Systems
AISTATS	Proceedings of the International Conference on Artificial Intelligence and Statistics
BBS	Behavioral and Brain Sciences
CACM	Communications of the Association for Computing Machinery
COGSCI	Proceedings of the Annual Conference of the Cognitive Science Society
COLING	Proceedings of the International Conference on Computational Linguistics
COLT	Proceedings of the Annual ACM Workshop on Computational Learning Theory
CP	Proceedings of the International Conference on Principles and Practice of Constraint Programming
CVPR	Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition
EC	Proceedings of the ACM Conference on Electronic Commerce
ECAI	Proceedings of the European Conference on Artificial Intelligence
ECCV	Proceedings of the European Conference on Computer Vision
ECML	Proceedings of the The European Conference on Machine Learning
ECP	Proceedings of the European Conference on Planning
EMNLP	Proceedings of the Conference on Empirical Methods in Natural Language Processing
FGCS	Proceedings of the International Conference on Fifth Generation Computer Systems
FOCS	Proceedings of the Annual Symposium on Foundations of Computer Science
GECCO	Proceedings of the Genetics and Evolutionary Computing Conference
HRI	Proceedings of the International Conference on Human-Robot Interaction
ICAPS	Proceedings of the International Conference on Automated Planning and Scheduling
ICASSP	Proceedings of the International Conference on Acoustics, Speech, and Signal Processing
ICCV	Proceedings of the International Conference on Computer Vision
ICLP	Proceedings of the International Conference on Logic Programming
ICLR	Proceedings of the International Conference on Learning Representations
ICML	Proceedings of the International Conference on Machine Learning
ICPR	Proceedings of the International Conference on Pattern Recognition
ICRA	Proceedings of the IEEE International Conference on Robotics and Automation
ICSLP	Proceedings of the International Conference on Speech and Language Processing
IJAR	International Journal of Approximate Reasoning
IJCAI	Proceedings of the International Joint Conference on Artificial Intelligence
IJCNN	Proceedings of the International Joint Conference on Neural Networks
IJCV	International Journal of Computer Vision
ILP	Proceedings of the International Workshop on Inductive Logic Programming
IROS	Proceedings of the International Conference on Intelligent Robots and Systems
ISMIS	Proceedings of the International Symposium on Methodologies for Intelligent Systems
ISRR	Proceedings of the International Symposium on Robotics Research
JACM	Journal of the Association for Computing Machinery
JAIR	Journal of Artificial Intelligence Research
JAR	Journal of Automated Reasoning
JASA	Journal of the American Statistical Association
JMLR	Journal of Machine Learning Research
JSL	Journal of Symbolic Logic
KDD	Proceedings of the International Conference on Knowledge Discovery and Data Mining
KR	Proceedings of the International Conference on Principles of Knowledge Representation and Reasoning
LICS	Proceedings of the IEEE Symposium on Logic in Computer Science
NeurIPS	Advances in Neural Information Processing Systems
PAMI	IEEE Transactions on Pattern Analysis and Machine Intelligence
PNAS	Proceedings of the National Academy of Sciences of the United States of America
PODS	Proceedings of the ACM International Symposium on Principles of Database Systems
RSS	Proceedings of the Conference on Robotics: Science and Systems
SIGIR	Proceedings of the Special Interest Group on Information Retrieval
SIGMOD	Proceedings of the ACM SIGMOD International Conference on Management of Data
SODA	Proceedings of the Annual ACM-SIAM Symposium on Discrete Algorithms
STOC	Proceedings of the Annual ACM Symposium on Theory of Computing
TARK	Proceedings of the Conference on Theoretical Aspects of Reasoning about Knowledge
UAI	Proceedings of the Conference on Uncertainty in Artificial Intelligence

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α (alpha) learning rate, 816
 α (alpha) normalization constant, 418
 $\wedge$ (and), 235
 χ^2 (chi squared), 682
 $\vdash$ (derives), 234
 $\models$ (entailment), 232
 ε (epsilon)-ball, **691**
 $\exists$ (there exists), 280
 $\forall$ (for all), 279
 γ (gamma) discount factor, 632
 $\neg$ (gap) in sentence, 897
 $|$ (given), 407
 $\Leftrightarrow$ (if and only if), 235
 $\Rightarrow$ (implies), 235
 $\sim$ (indifferent), 520
 λ (lambda)-expression, **277**
 ∇ (nabla) gradient, 138
 $\neg$ (not), 235
 $\vee$ (or), 235
 $\succ$ (preferred), 520
 σ (sigma) standard deviation, 1079
 T (matrix transpose), 1077

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